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AIM:

TASKS A, B, C, D, E

Q1] Scan a given network range and identify all active hosts.

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 11:41 +0530
Nmap scan report for 192.168.1.2
Host is up (0.022s latency).
Nmap done: 256 IP addresses (1 host up) scanned in 23.10 seconds
```

Ans]

A Ping Scan (-sn) in Nmap is used to discover active hosts on a network without performing a port scan. It sends ICMP echo requests and other probe packets to determine which devices are online and reachable.

Q2] Identify the top 5 most commonly open ports on a specific target.

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap --top-ports 5 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 11:53 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
21/tcp    closed ftp
22/tcp    open  ssh
23/tcp    closed telnet
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```



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Nmap can target only the most commonly used ports instead of scanning all 65,535 ports, enabling quicker detection of frequently used network services.

Q3] Determine the MAC address of a target device using Nmap.

Proof:

```
[siddh.s7@Siddhs-MacBook-Air ~ % sudo nmap -sn 192.168.1.1  
[Password:  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 11:58 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds
```

ANS]

The MAC address uniquely identifies a device on a local network. Nmap can discover the MAC address of hosts during a ping scan on the local network.

Q4] Detect the presence of HTTP and HTTPS services on a target network.

Proof:

```
[siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 80,443 scanme.nmap.org  
  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 11:58 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.24s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   closed https  
  
Nmap done: 1 IP address (1 host up) scanned in 0.53 seconds
```

ANS]

HTTP and HTTPS are web services that run on port 80 and port 443 respectively. Nmap can scan specific ports to determine whether these services are active.

Q5] Find out if a particular host has FTP service running

Proof:



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```
[siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 21 scanme.nmap.org ]
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:00 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
21/tcp    closed ftp

Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds
```

ANS]

FTP is a protocol used for file transfer and typically runs on port 21. Nmap can scan this port to check whether the FTP service is active.

Q6] Identify the SSH version running on a given host.

Proof:

```
[siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 22 -sV scanme.nmap.org ]
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:01 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.40 seconds
```

ANS]

SSH is used for secure remote login and operates on port 22. Using service version detection, Nmap can identify the software and version running on a port.

Q7] Scan a range of IP addresses and list hosts that have Telnet service running

Proof:



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```
[siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 23 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:04 +0530
Nmap scan report for 192.168.1.2
Host is up (0.024s latency).

PORT      STATE SERVICE
23/tcp    closed telnet

Nmap done: 256 IP addresses (1 host up) scanned in 22.37 seconds
```

ANS]

Telnet is a remote login protocol that runs on port 23. Scanning a network range helps identify devices offering Telnet services.

Q8]Determine the operating system of a target host using Nmap.

Proof:

```
[siddh.s7@Siddhs-MacBook-Air ~ % sudo nmap -O scanme.nmap.org
[Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:06 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
445/tcp    filtered microsoft-ds
1080/tcp   filtered socks
2000/tcp   open  cisco-sccp
5060/tcp   open  sip
9929/tcp   open  nping-echo
31337/tcp  open  Elite
Device type: general purpose
Running: Linux 4.X
OS CPE: cpe:/o:linux:linux_kernel:4
OS details: Linux 4.19 – 5.15
Network Distance: 24 hops

OS detection performed. Please report any incorrect results at https://nmap.org/sub
mit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.05 seconds
```

ANS]

Nmap can detect the operating system of a host by analyzing TCP/IP stack characteristics. This



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technique is known as OS fingerprinting.

Q9] Identify any SQL services running on a network.

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 1433,3306 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:07 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.33s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
1433/tcp  closed ms-sql-s
3306/tcp  closed mysql

Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds
```

ANS]

Database services such as MySQL and Microsoft SQL Server run on specific ports. Scanning these ports allows detection of database servers in a network.

Q10] Find out if a specific host has Remote Desktop Protocol (RDP) enabled.

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 3389 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:11 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
3389/tcp  closed ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.90 seconds
```

ANS]

Remote Desktop Protocol allows remote access to Windows systems and runs on port 3389. Nmap can scan this port to determine if remote desktop services are available.

Q11] Scan a network and determine if hosts are running DNS services.

Proof:



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```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 53 scanme.nmap.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:12 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
53/tcp    closed domain

Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds
```

ANS:

DNS is responsible for translating domain names into IP addresses and runs on port 53. Nmap can detect DNS servers by scanning this port.

Q12] Detect if a host has SNMP enabled

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 161 scanme.nmap.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:13 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
161/tcp    closed snmp

Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds
```

ANS:

SNMP is used for monitoring and managing network devices and operates on port 161. Scanning this port helps identify devices with SNMP services.

Q13] Identify any SMTP servers on a network.

Proof:



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```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 25 scanme.nmap.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:14 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
25/tcp    closed smtp

Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds
```

ANS]

SMTP is the protocol used for sending emails and typically runs on port 25. Nmap can scan this port to check whether an SMTP server is active.

Q14] Determine if FTP servers allow anonymous login.

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap --script ftp-anon -p 21 scanme.nmap.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:15 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
21/tcp    closed ftp

Nmap done: 1 IP address (1 host up) scanned in 1.29 seconds
```

ANS]

Some FTP servers allow users to log in anonymously without authentication. Nmap scripts can test whether anonymous FTP access is permitted.

Q15] Find hosts running vulnerable versions of the Apache HTTP server.

Proof:



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```
[siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 80 --script http-enum scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:36 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 52.35 seconds
siddh.s7@Siddhs-MacBook-Air ~ %
```

ANS:

Apache is a widely used web server, but certain versions may contain vulnerabilities. Nmap vulnerability scripts can check web servers for known issues.

Q16] Detect if a host has NFS shares available

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 2049 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:16 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
2049/tcp  closed nfs

Nmap done: 1 IP address (1 host up) scanned in 0.60 seconds
```

ANS:

NFS (Network File System) allows file sharing across networks and typically runs on port 2049. Nmap can scan this port to identify NFS services.

Q17] Identify the presence of MySQL database servers.

Proof:



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```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 3306 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:17 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
3306/tcp   closed mysql

Nmap done: 1 IP address (1 host up) scanned in 0.61 seconds
```

ANS:

MySQL is a database service that runs on port 3306. Scanning this port helps locate MySQL servers within a network.

Q18] Detect hosts running Remote Procedure Call (RPC) services.

Proof:

```
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 111 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:18 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
111/tcp    closed rpcbind

Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
```

ANS:

PC allows programs to execute procedures on remote systems and commonly uses port 111. Scanning this port reveals hosts running RPC services.

Q19] Detect if a host has VNC (Virtual Network Computing) enabled.

Proof:



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```
nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 5900 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:19 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
5900/tcp  closed vnc

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

ANS:

VNC allows remote graphical access to a system and typically runs on port 5900. Nmap can scan this port to identify systems offering VNC services.

Q20] Identify hosts running SSH on non-default ports.

Proof:

```
[siddh.s7@Siddhs-MacBook-Air ~ % nmap -p 1-2000 -sV scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-04 12:32 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 1997 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
2000/tcp  open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 139.70 seconds
```

ANS:

SSH normally runs on port 22, but administrators may configure it on different ports for security. Scanning all ports with service detection can reveal SSH services running on non-standard ports.

Conclusion:

In this experiment, we used Nmap to perform systematic network scanning and service enumeration on a target host. Different scanning techniques were applied to identify open ports, detect running services, determine service versions, and perform operating system fingerprinting. Through targeted port scans and service detection options, we successfully identified common network services such as HTTP, HTTPS, FTP, SSH, Telnet, DNS, SMTP, MySQL, SNMP, RPC,



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RDP, VNC, and NFS. We also utilized NSE (Nmap Scripting Engine) scripts to check for vulnerabilities and anonymous FTP access. Additionally, OS detection and version detection techniques demonstrated how Nmap analyzes TCP/IP stack behavior to fingerprint operating systems and running software. Overall, this experiment provided practical exposure to network reconnaissance techniques, enhanced understanding of common network protocols and their default ports, and demonstrated how Nmap can be effectively used for security assessment and vulnerability identification in a controlled environment.